LESSON PLAN

GAS TURBINE ENGINE (16)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Inspection of neutron radiography introduction, different methods principle and various applications
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Inspection of neutron radiography introduction, different methods principle and various applications
5. Specific Objectives	16.1.3.1 Trainee will be able to explain the principle of radiographic NDT.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of radiographic NDT principles using PPT slides; Instructor supervision of trainee note-taking and Q&A; session.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary difference between X-ray and neutron radiography?
(b) Name one material that is difficult to inspect using X-rays but easy with neutrons.
(c) Define the term 'radiographic NDT'.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Neutron Radiography • Principle of Radiographic NDT • Interaction of Neutrons with Matter • Neutron Source Types and Selection • Image Formation and Detection Methods	L & Ds	PPT, IB	15 min
(b)	• Different Methods of Neutron Radiography • Transmission vs. Backscatter Techniques • Fission vs. Radioisotope Sources • Digital vs. Film Detection Systems • Safety Precautions for Neutron Fields	L & Ds	PPT, PDF	20 min
(c)	• Various Applications in Aerospace • Inspection of Composite Materials • Detection of Hydrogenous Defects • Corrosion Assessment in Reactors • Limitations and Future Trends	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the principle of radiographic NDT applies to neutron imaging.
(b) Describe one specific application of neutron radiography in aircraft maintenance.

12. Home Assignment and Follow Up	Prepare a brief note comparing the advantages of neutron radiography over ultrasonic inspection for detecting defects in hydrogen-rich composite structures, referencing concepts from LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Ultra sonic inspection, basic principle waves propagation characteristics acoustic impedance basic inspection methods, ultrasonic equipment, data representation, transducer, various probe couplant	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min